

Presentation Script for Cogs 105

From Checklists to Chemistry, What Predicts a Match?

Slide 1: Title Slide

Hello, my name is Cecilia Kuang and this is my final project for Cogs 105. This project looks at a pretty relatable question: do we actually know what we want in a partner, or do our choices end up being driven by something else? I'm going to be talking about what actually predicts romantic matches in speed dating: people's stated preferences, or the chemistry that happens during the date.

Slide 2: What are the questions?

This is the main question I explore throughout this project: Can what people say they want in a partner actually predict who they end up matching with? So can pre-date theories about partner preferences predict mutual romantic attraction in speed dating?

Before these speed dates happened, participants reported what they thought mattered most to them in a partner – things like attractiveness, sincerity, intelligence, fun, ambition, and shared interests. They even had to allocate points out of 100 across these traits to reflect their priorities. After the dates, they would rate their actual impressions of their partners on those same traits, which allowed us to compare what people say they value with what actually predicts mutual attraction. In theory, if people really know what they want, those preferences should help explain who they choose to match with (they would be the strongest predictors).

Slide 3: What are the results?

What did I discover? It was actually very interesting. People often say they know what they want in a partner, but their match decisions are driven more by how a date actually feels than by what they thought mattered beforehand.

I came to this conclusion from my model results. I tested three different types of models. First, Model A, which used demographics like age and race. That model performed pretty poorly. It mostly predicted 'no match' for everyone, which inflated accuracy but

didn't really capture attraction. So demographics alone don't tell us much about romantic interest. Next was Model B, which used people's pre-date stated preferences. Basically, what they said mattered to them before the event. Things like how much they value attractiveness, fun, intelligence, sincerity, ambition, and shared interests. This model did have some predictive power, but it was moderate and unstable. In other words, what people say they care about does help a little, but it's not very reliable for predicting real matches. The strongest results came from Model C, which used post-date ratings: how participants actually felt about their partner after interacting with them. These models consistently achieved higher predictive accuracy and ROC AUC. This tells us that what happens during the interaction matters much more than what people think matters beforehand.

The bar graph here shows permutation importance for Model C. Each bar represents how much prediction accuracy drops when we shuffle that variable. Higher bars mean the feature is more important. What we see is that post-date impressions like attractiveness, shared interests, and partner-perceived ambition are the most influential predictors of a match. These are not abstract theories, they reflect lived experience during the date. Taken together, these results show a clear pattern: People believe they know what they want, but their actual dating choices are driven more by how the interaction feels in the moment. This reveals a metacognitive gap between stated preferences and real behavior. The results suggest that people's abstract beliefs about what they value in a partner do not fully align with the factors that influence their real-world dating decisions.

Slide 4: What did you learn? What are the answers?

So what do we actually know now that we didn't before?

The biggest takeaway from this project is that people are not very good at predicting what will drive their romantic decisions. Before dating, participants had clear ideas about what they thought they valued in a partner—things like sincerity, ambition, or intelligence. But when we looked at what actually predicted mutual matches, those stated preferences were surprisingly weak. Instead, what mattered most were the impressions formed during the interaction itself. Real-time experiences, especially how attractive someone felt, whether they seemed fun, whether they shared interests, and how ambitious they came across, were far more informative than pre-date theories.

This suggests that our conscious dating preferences often function more like explanations we tell ourselves rather than accurate predictors of our behavior. Attraction seems to emerge during the interaction, not from a checklist we carry into it.

In other words, people may think they know what they want, but real romantic choice is shaped by how a date actually feels in the moment. Real attraction emerges during interaction, not before it.